Exercise 1 – Code Tracing

Exercise 1.1

a) Consider the following code. What is the output of the program?

```cpp
#include <iostream>

int main() {
    int i = 6;
    while (i > 0) {
        if (i == 4) {
            i = 2;
            continue;
        }
        if (i % 2 == 0) {
            std::cout << "Hello ";
        } else {
            std::cout << "there ";
        }
        i = i - 1;
    }
}
```

i	Output	Iteration

Output: _____

b) Which statement about the above program is correct?

- Because of the missing return statement, the main function automatically returns 1.
- Because of the missing return statement, the main function does not return a value.
- Because of the missing return statement, the code does not compile.
- The variable *i* has never the value 3.

Exercise 1.2

a) Consider the following code. What is the output of the program?

```cpp
#include <iostream>
#include <string>

using namespace std;

char someFunc() { return '\"'; }

string otherFunc() { return "nothing"; }

int main() {
  bool execute = false;
  string ObiWan{"\"I have the highground."};

  for (int i{0}; i < 8; ++i) {
    cout << ObiWan.at(i);
  }

  do {
    cout << "something";
  } while (execute);

  while (execute) {
    cout << otherFunc();
  }

  cout << someFunc() << endl;
  return 0;
}
```

i	Output	Iteration

Output: _____

b) Which statement about the above program is correct?

- The program contains several infinite loops.
- The strings is ouputed character by character.
- The program contains only two functions.
- The variable *i* of the for-loop can be used in the following do-while-loop. The variable *i* has the value 8.

Exercise 1.3

a) Consider the following code. What is the output of the program?

```
1   #include <iostream>
2
3   using namespace std;
4
5   int someFunc(int a, int b){
6       while(a > 0){
7           if(a < b){
8               int temp {a};
9               a = b;
10              b = temp;
11          }
12          a = a - b;
13      }
14      return b;
15  }
16
17  int main(){
18      int a {36};
19      int b {48};
20
21      cout << someFunc(42, a) <<endl;
22  }
```

a	b	Iteration

Output:_____

b) What best describes line 7 to 11?

- The variable *temp* is a combination of the variables *a* and *b*.
- A swap of the variables *a* and *b* is performed.
- The variable *temp* gets initialized to be used outside the if-Statement as an alternative for *a*.
- Something gets calculated with *a*, *b* and *temp*.

c) What best describes someFunc?

- The function calculates the greatest common divisor of a number.
- The function returns the smaller value of both numbers which are passed to it.
- The variable *a* is never 0.
- The function performs a prime factorization.